OV-based truly-subquadratic hardness of diameter at most three for unit-ball graphs in three dimensions

Timothy M. Chan, Hsien-Chih Chang, Jie Gao, Sándor Kisfaludi-Bak, Hung Le

doi:10.4230/LIPIcs.SoCG.2026.29; CC BY 4.0

Editorial scope. Complete official section2 and References only. Count Theorem9; preserve Observation11 and Lemma12 as support. Official algorithms of sections3/4 defer core details to NC full version and are excluded; shatter bounds not split. The reduction realizes a sparse tripartite graph by four layers of radius-1/2 balls in three dimensions. The author controls trigonometric coordinate gaps and squared distances so adjacent layers encode exactly the graph incidences, handles RAM precision, and proves the diameter-2/diameter-3 equivalence. This complete geometric encoding is the independent hardness result; no support observation or algorithm result is split.

Allowed standard prerequisites (editorial).

- Orthogonal Vectors hypothesis and the cited established OV lower bound for Diameter-2 on sparse tripartite graphs.. Official section2 explicitly calls the sparse tripartite Diameter-2 lower bound well-known and cites reference [19]; Lemma10 states the exact external bound. It remains an external prerequisite. All new four-layer coordinate and diameter equivalence proofs are fully included.

2 Diameter-3 for Unit Balls in 3D

Theorem 9.
Assuming the OV hypothesis, there is no $O(n^{2-\varepsilon})$ time algorithm for deciding if the intersection graph of a given set of n unit balls in $\mathbb{R}^3$ has diameter at most 3.

The reduction is from Diameter-2 for n-vertex sparse tripartite graphs $G = (A \cup B \cup C, E)$, where the number of edges m is $\tilde{O}(n)$. It is well-known that solving Diameter-2 for sparse tripartite graphs in truly-subquadratic time would break the OV hypothesis [19].

Lemma 10 (Diameter-2 sparse tripartite graphs).
There is no $O(n^{2-\varepsilon})$ time algorithm for deciding if a sparse tripartite graph has diameter at most 2, assuming the OV hypothesis.

In this section, we work with balls of radius $1/2$ instead of 1. Let $\varepsilon = \Theta(1/n^4)$ and $\delta = \Theta(1/n^3)$. First, we map every vertex v in $A \cup C$ to a distinct number, also denoted by v, in $[\varepsilon, 2\varepsilon]$, such that $\min_{a_1,a_2 \in A:\, a_1 \neq a_2} |a_1 - a_2| \geq \Omega(\varepsilon/n)$ and $\min_{c_1,c_2 \in C:\, c_1 \neq c_2} |c_1 - c_2| \geq \Omega(\varepsilon/n)$. Vertices of B are mapped to distinct numbers in $[0.9, 1]$ such that $\min_{b_1,b_2 \in B:\, b_1 \neq b_2} |b_1 - b_2| \geq \Omega(1/n)$. Then we create four sets of balls as follows.

1. For every vertex $a \in A$, we add a ball centered at $s_a = (1 + a\cos\delta, a\sin\delta, 0)$.
2. For every edge $(a, b) \in (A \times B) \cap E$, we add a ball centered at $p_{ab} = (1 + a\cos\delta - b\sin\delta, a\sin\delta + b\cos\delta, \sqrt{1 - b^2})$.
3. For every edge $(b, c) \in (B \times C) \cap E$, we add a ball centered at $q_{bc} = (-c, b, \sqrt{1 - b^2})$.
4. For every vertex $c \in C$, we add a ball centered at $t_c = (-c, 0, 0)$.

Let $\mathcal{B}_i$ be the set of balls created in step i for $i \in [4]$ and $\mathcal{B} \coloneqq \cup_i \mathcal{B}_i$. Observe that:

Observation 11.
Let $(\!|p|\!)$ be the ball of radius $1/2$ centered at a point p. We have:

1. $(\!|s_a|\!) \cap (\!|p_{\tilde{a}b}|\!) \neq \emptyset$ if and only if $a = \tilde{a}$.
2. $(\!|p_{ab}|\!) \cap (\!|q_{\tilde{b}c}|\!) \neq \emptyset$ if and only if $b = \tilde{b}$.
3. $(\!|p_{bc}|\!) \cap (\!|t_{\tilde{c}}|\!) \neq \emptyset$ if and only if $c = \tilde{c}$.
4. Every two balls in $\mathcal{B}_i$ intersect for every $i \in [4]$.
5. For any two balls $b_1 \in \mathcal{B}_i$ and $b_2 \in \mathcal{B}_j$ such that $|i - j| \geq 2$, $b_1 \cap b_2 = \emptyset$.

Proof.
For item 1, observe that

$$\begin{aligned}\|s_a - p_{\tilde{a}b}\|_2^2 &= ((a - \tilde{a})\cos\delta + b\sin\delta)^2 + ((a - \tilde{a})\sin\delta - b\cos\delta)^2 + 1 - b^2 \\ &= (a - \tilde{a})^2(\cos^2\delta + \sin^2\delta) + b^2(\sin^2\delta + \cos^2\delta) + 1 - b^2 \;=\; (a - \tilde{a})^2 + 1,\end{aligned}$$

which is exactly 1 if $a = \tilde{a}$, and at least $1 + \Omega(1/n^{10})$ if $|a - \tilde{a}| \geq \Omega(\varepsilon/n)$.
For item 2, observe that

$$\begin{aligned}&\|p_{ab} - q_{\tilde{b}c}\|_2^2 \\ =&(1 + a\cos\delta - b\sin\delta + c)^2 + (a\sin\delta + b\cos\delta - \tilde{b})^2 + (\sqrt{1 - b^2} - \sqrt{1 - \tilde{b}^2})^2 \\ =&(1 + \Theta(\varepsilon) - \Theta(\delta) + \Theta(\varepsilon))^2 + (\Theta(\varepsilon\delta) + b - \Theta(\delta^2) - \tilde{b})^2 + (\sqrt{1 - b^2} - \sqrt{1 - \tilde{b}^2})^2 \\ =&(1 - \Theta(1/n^3))^2 + (|b - \tilde{b}| \pm \Theta(1/n^6))^2 + (\sqrt{1 - b^2} - \sqrt{1 - \tilde{b}^2})^2,\end{aligned}$$

which is $1 - \Theta(1/n^3)$ if $b = \tilde{b}$, and at least $1 + \Omega(1/n^2)$ if $|b - \tilde{b}| \geq \Omega(1/n)$.

For item 3, observe that $\|q_{bc} - t_{\tilde{c}}\|_2^2 = (\tilde{c} - c)^2 + 1$, which is exactly 1 if $c = \tilde{c}$, and at least $1 + \Omega(1/n^{10})$ if $|c - \tilde{c}| \geq \Omega(\varepsilon/n)$. Item 4 is easy to check. For item 5, observe that $\|s_a - t_c\|_2 \geq 1 + a\cos\delta + c \geq 1 + \Omega(1/n^4)$, whereas $\|s_a - q_{bc}\|_2$ and $\|p_{ab} - t_c\|_2$ are both $\sqrt{2} \pm o(1)$. ◀

In the real RAM model, balls $\mathcal{B}$ can clearly be constructed in $O(n + m) = \tilde{O}(n)$ time. In the RAM model, it suffices to approximate all coordinates up to $O(1/n^{10})$ additive error, which can be done in $\tilde{O}(1)$ time (we could actually pick "nice" choices of $b \in B$ so that b and $\sqrt{1 - b^2}$ are both rational, and also a nice δ so that $\cos\delta$ and $\sin\delta$ are both rational). Thus, the reduction can be done in $\tilde{O}(n)$ time. Theorem 9 follows from the following lemma.

Lemma 12.

Let K be the intersection graph of the set of balls $\mathcal{B}$. Then G has diameter at most 2 if and only if K has diameter at most 3.

Proof.

Observe by item 4 of Observation 11 that the balls in the same set $\mathcal{B}_i$ induce a clique in K. By item 5 of Observation 11, there is no edge between two balls in $\mathcal{B}_i$ and $\mathcal{B}_j$ when $|j - i| \geq 2$. Thus, K has diameter at most 3 if and only if for every ball $[\![s_a]\!] \in \mathcal{B}_1$ and every ball $[\![t_c]\!] \in \mathcal{B}_4$, $d_K(s_a, t_c) \leq 3$. Here we slightly abuse the notation by using the center of the ball to denote the corresponding vertex in K.

Suppose that G has diameter at most 2. Let $P = (a, b, c)$ be any path of length 2 in G. Then by Observation 11, $(s_a, p_{ab}, q_{bc}, t_c)$ is a path of length 3 in K. Thus, K has diameter at most 3 by the above observation. For the other direction, if K has diameter at most 3, then for any two vertices s_a and t_c (such that $[\![s_a]\!] \in \mathcal{B}_1$ and $[\![t_c]\!] \in \mathcal{B}_4$), a path of length 3 between s_a and t_c in K must be of the form $(s_a, p_{\tilde{a}b}, q_{\tilde{b}c}, t_c)$. Then again by Observation 11, $a = \tilde{a}$ and $b = \tilde{b}$, and therefore, (a, b, c) is a path of length 2 in G. ◀

References

[1] A. Karim Abu-Affash, Paz Carmi, Anil Maheshwari, Pat Morin, Michiel Smid, and Shakhar Smorodinsky. Approximating maximum diameter-bounded subgraph in unit disk graphs. Discret. Comput. Geom., 66(4):1401–1414, 2021. doi:10.1007/s00454-021-00327-y.

[2] Pankaj K Agarwal. Range searching. In Handbook of discrete and computational geometry, pages 1057–1092. Chapman and Hall/CRC, 2017. doi:10.1201/9781315119601.

[3] Sujoy Bhore, Timothy M. Chan, Zhengcheng Huang, Shakhar Smorodinsky, and Csaba D. Tóth. Sparse bounded hop-spanners for geometric intersection graphs. In Oswin Aichholzer and Haitao Wang, editors, 41st International Symposium on Computational Geometry, SoCG 2025, Kanazawa, Japan, June 23-27, 2025, volume 332 of LIPIcs, pages 17:1–17:15. Schloss Dagstuhl – Leibniz-Zentrum für Informatik, 2025. doi:10.4230/LIPIcs.SOCG.2025.17.

[4] Nicolas Bousquet and Stéphan Thomassé. VC-dimension and Erdős–Pósa property. Discret. Math., 338(12):2302–2317, 2015. doi:10.1016/j.disc.2015.05.026.

[5] Karl Bringmann, Sándor Kisfaludi-Bak, Marvin Künnemann, André Nusser, and Zahra Parsaeian. Towards sub-quadratic diameter computation in geometric intersection graphs. In Xavier Goaoc and Michael Kerber, editors, 38th International Symposium on Computational Geometry, SoCG 2022, Berlin, Germany, June 7-10, 2022, volume 224, pages 21:1–21:16. Schloss Dagstuhl – Leibniz-Zentrum für Informatik, 2022. doi:10.4230/LIPIcs.SoCG.2022.21.

[6] Sergio Cabello. Subquadratic algorithms for the diameter and the sum of pairwise distances in planar graphs. ACM Trans. Algorithms, 15(2):21:1–21:38, December 2019. doi:10.1145/3218821.

[7] Timothy M. Chan. Finding triangles and other small subgraphs in geometric intersection graphs. In Nikhil Bansal and Viswanath Nagarajan, editors, Proceedings of the 2023 ACM-SIAM Symposium on Discrete Algorithms, SODA 2023, Florence, Italy, January 22-25, 2023, pages 1777–1805. SIAM, 2023. doi:10.1137/1.9781611977554.CH68.

[8] Timothy M. Chan, Hsien-Chih Chang, Jie Gao, Sándor Kisfaludi-Bak, Hung Le, and Da Wei Zheng. Truly subquadratic time algorithms for diameter and related problems in graphs of bounded VC-dimension. In 66th Annual IEEE Symposium on Foundations of Computer Science (FOCS), pages 2728–2765, 2025. doi:10.48550/arXiv.2510.16346.

[9] Timothy M. Chan, Hsien-Chih Chang, Jie Gao, Sándor Kisfaludi-Bak, Hung Le, and Da Wei Zheng. Charting the diameter computation landscape on intersection graphs in 2D, 2026. Under preparation.

[10] Timothy M. Chan, Qizheng He, and Yuancheng Yu. On the fine-grained complexity of small-size geometric set cover and discrete k-center for small k. In Kousha Etessami, Uriel Feige, and Gabriele Puppis, editors, 50th International Colloquium on Automata, Languages, and Programming, ICALP 2023, Paderborn, Germany, July 10-14, 2023, volume 261 of LIPIcs, pages 34:1–34:19. Schloss Dagstuhl – Leibniz-Zentrum für Informatik, 2023. doi:10.4230/LIPIcs.ICALP.2023.34.

[11] Hsien-Chih Chang, Jie Gao, and Hung Le. Computing diameter+2 in truly-subquadratic time for unit-disk graphs. In Wolfgang Mulzer and Jeff M. Phillips, editors, 40th International Symposium on Computational Geometry, SoCG 2024, Athens, Greece, June 11-14, 2024, volume 293, pages

38:1–38:14. Schloss Dagstuhl – Leibniz-Zentrum für Informatik, 2024. Arxiv version https://arxiv.org/abs/2401.12881. doi:10.4230/LIPIcs.SoCG.2024.38.

[12] Victor Chepoi, Bertrand Estellon, and Yann Vaxès. Covering planar graphs with a fixed number of balls. Discret. Comput. Geom., 37(2):237–244, 2007. doi:10.1007/s00454-006-1260-0.

[13] Guillaume Ducoffe, Michel Habib, and Laurent Viennot. Diameter, eccentricities and distance oracle computations on H-minor free graphs and graphs of bounded (distance) vapnik-chervonenkis dimension. SIAM J. Comput., 51(5):1506–1534, 2022. doi:10.1137/20M136551X.

[14] Lech Duraj, Filip Konieczny, and Krzysztof Potępa. Better diameter algorithms for bounded VC-dimension graphs and geometric intersection graphs. In Timothy M. Chan, Johannes Fischer, John Iacono, and Grzegorz Herman, editors, 32nd Annual European Symposium on Algorithms, ESA 2024, Royal Holloway, London, United Kingdom, September 2-4, 2024, volume 308, pages 51:1–51:18. Schloss Dagstuhl – Leibniz-Zentrum für Informatik, 2024. doi:10.4230/LIPIcs.ESA.2024.51.

[15] Paweł Gawrychowski, Haim Kaplan, Shay Mozes, Micha Sharir, and Oren Weimann. Voronoi diagrams on planar graphs, and computing the diameter in deterministic $\tilde{O}(n^{5/3})$ time. SIAM J. Comput., 50(2):509–554, 2021. doi:10.1137/18M1193402.

[16] Hung Le and Christian Wulff-Nilsen. VC set systems in minor-free (di)graphs and applications. In David P. Woodruff, editor, Proceedings of the 2024 ACM-SIAM Symposium on Discrete Algorithms, SODA 2024, Alexandria, VA, USA, January 7-10, 2024, pages 5332–5360. SIAM, 2024. doi:10.1137/1.9781611977912.192.

[17] Jason Li and Merav Parter. Planar diameter via metric compression. In Moses Charikar and Edith Cohen, editors, Proceedings of the 51st Annual ACM SIGACT Symposium on Theory of Computing, STOC 2019, Phoenix, AZ, USA, June 23-26, 2019, pages 152–163, Phoenix, AZ, USA and New York, NY, USA, 2019. ACM. doi:10.1145/3313276.3316358.

[18] Jiří Matoušek. Range searching with efficient hierarchical cuttings. Discret. Comput. Geom., 10:157–182, 1993. doi:10.1007/BF02573972.

[19] Liam Roditty and Virginia Vassilevska Williams. Fast approximation algorithms for the diameter and radius of sparse graphs. In Dan Boneh, Tim Roughgarden, and Joan Feigenbaum, editors, Symposium on Theory of Computing Conference, STOC'13, Palo Alto, CA, USA, June 1-4, 2013, pages 515–524. ACM, 2013. doi:10.1145/2488608.2488673.

[20] Norbert Sauer. On the density of families of sets. J. Comb. Theory A, 13(1):145–147, 1972. doi:10.1016/0097-3165(72)90019-2.

[21] Saharon Shelah. A combinatorial problem; stability and order for models and theories in infinitary languages. Pacific Journal of Mathematics, 41(1):247–261, 1972. doi:10.2140/pjm.1972.41.247.